Laboratory work 11

1 Objectives

This laboratory presents the key notions on Bezier curves.

2 Theoretical background

2.1 Bezier curves

A Bezier curve is a parametric curve, used in computer graphics and other related fields, used to model smooth curves that can be scaled indefinitely. Another applicability of the Bezier curves is in animations where an object movement can be defined by using a Bezier curve, modifying in this way the velocity of the object.

2.2 Cubic Bezier curves

In order to define a cubic Bezier curve we need 4 points. The curve will pass through P0 and P3, which are the starting point and the ending point of the curve. The curve will not pass through P1 and P3 which are control points and are used to provide directional information.

$$\boldsymbol{B}(t) = (1-t)^3\boldsymbol{P}_0 + 3(1-t)^2t\boldsymbol{P}_1 + 3(1-t)t^2\boldsymbol{P}_2 + t^3\boldsymbol{P}_3\,, t \in [0,1]$$

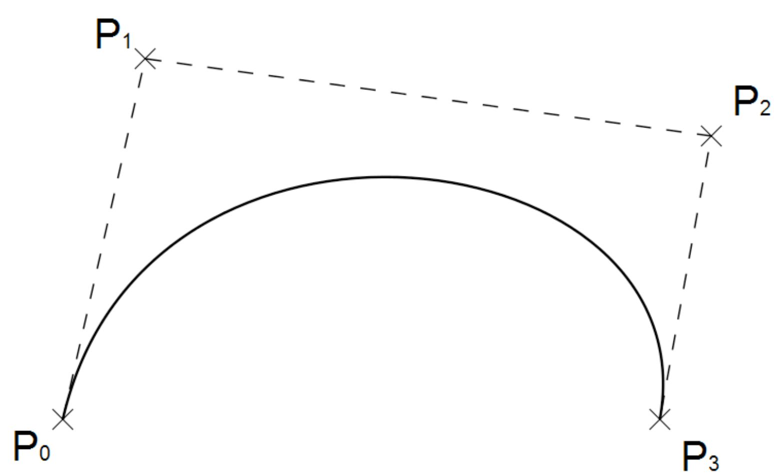

Fig. 1: Example of a Bezier curve with 4 points

2.3 Generalization

The Bezier curve of degree n can be generalized in this way:

$$B(t) = \sum_{i=0}^{n} \binom{n}{i} (1-t)^{n-i} t^i P_i$$

For example, for n = 5:

$$B(t) = (1-t)^5 P_0 + 5t(1-t)^4 P_1 + 10t^2(1-t)^3 P_2 + 10t^3(1-t)^2 P_3 + 5t^4(1-t) P_4 + t^5 P_5 ,$$

$$t \in [0,1]$$

Recursively the formula can be expressed as follows:

$$B(t) = B_{P0P1\ldots Pn}(t) = (1-t)B_{P0P1\ldots Pn-1}(t) + tB_{P1P2\ldots Pn}(t)$$

3 Assignment

- Create an application to exemplify the Bezier curves.